

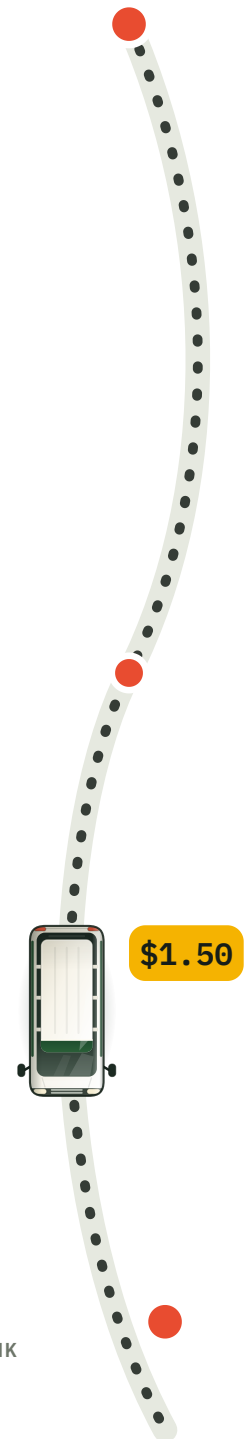


AI FOR IMPACT CHALLENGE 2026
TRACK 3, DEVELOPMENT
TRANSPORT, LOGISTICS AND MOBILITY

MOUNT PLEASANT
HEIGHTS

TRUSTED KOMBI TRAVEL FOR HARARE

**Safety, trust and
predictability for Harare's
kombi riders, and a record
the network has never
kept.**



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TEAM

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Section 1: Problem definition and strategic alignment

The problem: Harare's kombi network is unsafe, and no one is accountable

Half of Harare gets to work in a kombi. The city's own master plan puts about half of residents on kombis, buses and pirate taxis for the daily commute (Harare Master Plan 2025 to 2045). That network moves the city, and it is dangerous in a way that is now measurable.

Zimbabwe loses about 2,000 people on its roads every year. FactCheckZW rated that figure true against ZimStat and Zimbabwe Republic Police data, and the Traffic Safety Council of Zimbabwe records 52,288 crashes and 2,015 deaths for 2024 alone. The trend is worsening, not settling: road deaths in the first half of 2025 rose to 1,088, up from 1,037 in the same half the year before (ZRP, via The Herald). The Traffic Safety Council attributes 94 percent of crashes to human error (TSCZ, via The Herald and Africanews). The World Health Organization estimates Zimbabwe's road death rate at about 41 per 100,000 people, among the highest in Africa; the officially registered count is lower, nearer 30 per 100,000, and the WHO judges the real toll higher because many road deaths are never recorded (WHO, Global status report on road safety). In Harare specifically, recorded accidents jumped from 12,089 in 2021 to 22,670 in 2022 (Harare Master Plan).

Behind the numbers is a network that keeps no record of itself. Passengers board and step off at random points, pay cash, and get no ticket. Kombis carry no passenger manifest, so no one holds a record of who is aboard any vehicle at any moment (Gemini assessment and primary research, section 4.1). That gap turns an ordinary crash into a search. When a minibus and a haulage truck collided head on near Chitungwiza in July 2025 and 17 people died at the scene, police could only appeal to families to come to the mortuary and identify the remains themselves (AP News and ZRP statement, July 2025). There was no list to check against, because the system never made one.

So the daily experience of a Harare commuter is not only the fare and the wait. It is boarding a vehicle that may not be registered, driven by someone the network cannot name, with no record that you were ever on it. About 13,500 of roughly 16,500 kombis in Harare are unregistered, with only about 2,950 officially licensed (ZRP crackdown figures, 2025). The trust deficit is rational. People ride because they have no alternative, not because the arrangement is safe.

The user: the everyday rider, and the owner who wants a clean record

Svika is passenger first. The primary user is the commuter who needs to get across Harare and wants to know, before and during the ride, that the trip is predictable and that someone would know they were on it. The second user is the kombi owner who wants an honest record of what their vehicles earn and carry. The conductor sits between them, clearing fares and earning a fair share on digital ones.

The promise Svika makes to them is safety, trust and predictability. A rider plans a trip in Shona or English, boards with a short code that ties them to a specific vehicle and route, and can share a live trip with someone who would notice if it went wrong. The digital ticket is the manifest the network has never had. Digital ticketing, board codes and wallet credit are the mechanisms that produce the record and the trust; they are not the pitch.

Strategic alignment: National AI Strategy, Pillar 3

Zimbabwe's National Artificial Intelligence Strategy 2026 to 2030, issued by the Ministry of ICT, Postal and Courier Services with POTRAZ, sets a national vision of "AI for Development" that elevates citizen dignity (National AI Strategy, page 17). Svika sits directly inside Pillar 3, AI adoption and service transformation, whose transport sub sector commits

the country to "strengthen fleet management" and "improve public transport services" through intelligent systems (National AI Strategy, pages 24 to 25).

The fit runs wider than one pillar. The strategy names informal sector workers as a target for financial inclusion and mobile money (page 24); Svika's wallet and conductor commission bring the informal kombi trade a digital record and a payment rail. The strategy commits to natural language processing in local languages and to AI "free from imported biases" aligned with Ubuntu (pages 9 and 32); Svika is bilingual Shona and English from the first screen, with Ndebele next so Bulawayo and the Ndebele speaking south are included, and with Zimbabwean voice guidance for low literacy and blind riders. A purpose built Shona model, trained on the ZCHPC national compute, is on the roadmap to replace the current machine drafted strings with real modern nuance. The strategy prioritises computational sovereignty and in country data control (Pillar 2, page 22); every Svika model runs server side, deployable to the ZCHPC national compute, with no rider inference sent off device and no dependence on a foreign vendor in the ride path.

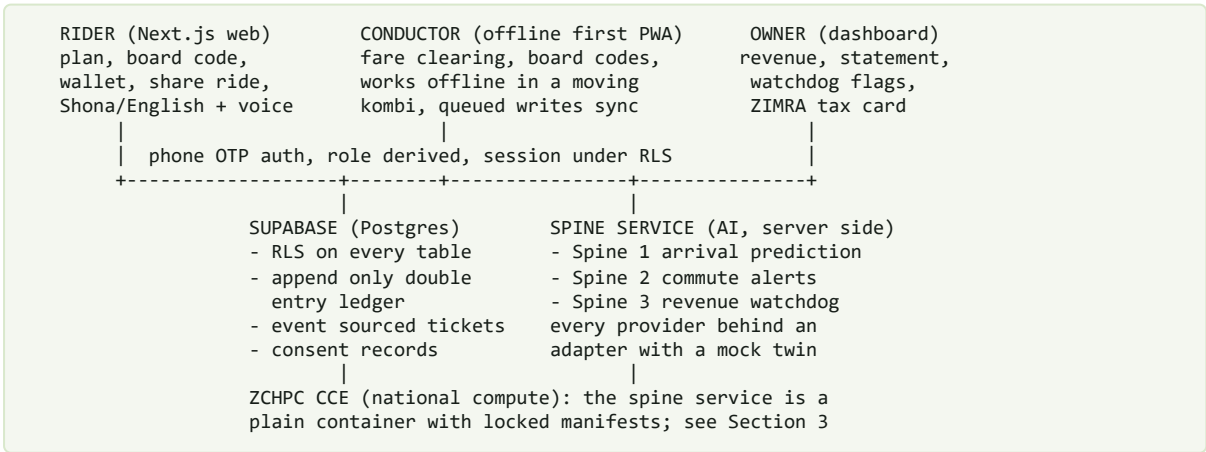
Beyond the rider, Svika's aggregated and consented movement data is a planning asset Harare has never had: because kombis thread the whole city, its main roads and its unmapped feeder routes alike, the network sees where people actually travel, where stops form and where a route or a lane is missing. In Nairobi, GPS data from the informal matatu network became the city's first complete transit map and entered Google Maps, a first for an informal system (Digital Matatus, MIT and the University of Nairobi); Svika can give Harare's municipalities the same, as anonymised aggregate insight under the consent and data protection rules in Section 4. That makes the city a second beneficiary and a partner, and it is a roadmap capability, not a feature we claim today.

Svika is not a general road safety programme, and it does not claim to be. It is a product that closes one specific, measurable gap: an informal network that carries half the city and keeps no record of who it carries. Closing that gap is where the safety, the trust and the national strategy meet.

Section 2: Technical design and product logic

System architecture

Svika is a monorepo of small, testable workspaces. Everything that touches money, tickets or rider identity lives behind Postgres row level security in a single Supabase project; every AI model runs server side in a separate spine service that never holds rider identity. Three surfaces sit in front of that: the rider web app, the conductor offline app, and the owner dashboard.



Two rules shape the whole design. Data integrity is enforced in the database, not in the application, so a bug in a screen cannot print money or leak a ticket. AI is isolated in one service behind adapters, so a model can be swapped or a vendor can go dark without touching the ride path.

Inference stays server side by product rule, a deliberate choice, not a gap: no model runs on a rider or conductor phone, which keeps riders' data off the device and on the ZCHPC national compute. The edge problem here is not model size on a handset, it is working when the network drops in a moving kombi, which Svika answers with the offline conductor app, not on device AI (Section 3). Section 5 carries the full edge and cost argument.

The data spine: integrity that a judge can test

Row level security is on every table from the first migration. The service role key exists only in the seed script and CI secrets, never in a running app. An automated security test proves, using the anonymous key alone, that one rider cannot read another rider's tickets or wallet; that suite now stands at 102 checks passing, including rider isolation, ledger safety and consent scoping (pnpm db:security-test, all green in CI).

Money is a double entry, append only ledger. There are no mutable balance columns. Every movement is a pair of postings that must sum to zero, and unit tests prove money cannot be created, lost or double spent before any wallet feature merges (ledger invariant tests, 8 passing, 35 transactions balance in the phase 1 gate). Unreturned change becomes wallet credit through the same ledger, so it is a posting like any other, not a special case that could leak.

Tickets are event sourced. A ticket's life is a sequence of appended `ticket_events` rows, never an update over history. This is what lets the digital ticket act as the manifest the network has never had: the record of who boarded which vehicle on which route cannot be quietly rewritten after the fact.

Offline is architecture, not a feature. The conductor app holds a local ticket cache in IndexedDB and clears fares with no signal, in a moving kombi, in sunlight. Queued writes reconcile on the next sync with a first sync wins rule and conflicts flagged, and the offline settlement path is the same code as the online one, proven by a live database cycle test that settles once, replays as a no op, and credits change exactly once (34 checks passing, back to back, proven by the committed offline test suite).

The three AI spines

Svika uses AI in exactly three places, and nowhere else. Each spine keeps a named baseline and a committed metrics file, and each serves the model only when it beats that baseline on held out data. Where a rule or a query is enough, Svika uses the rule or the query and says so. This is a direct answer to the rubric's justification audit: no forced AI, no sledgehammer.

Spine 1: arrival prediction with no timetable

The problem is real: the network has no schedule, so a rider cannot know when the next kombi arrives. The baseline is the naive per segment average. The model is a per segment, per hour average that can learn rush hour shape. The promotion rule is fixed in code: the model serves only when at least 10 journeys are recorded and it beats the baseline on a leave one journey out evaluation; serving code reads only the verdict in `metrics.json` (`services/spine/src/eta/train.ts`).

The data comes from routine travel: a custom field GPS logger turns an everyday commute along the Mount Pleasant Heights to Rezende corridor, a route the team already rides, into structured, reproducible segment data, and the training pipeline ingests every new ride

automatically. The model carries the discipline to defer to a transparent baseline until the data earns the switch. On two recorded journeys giving 20 segment observations, the baseline and the model tie at 96.4 seconds, below the ten journey bar, so the verdict is `insufficient_data` and `baseline:v1` serves (`services/spine/metrics/METRICS.md`, generated by `pnpm spine:train`). The promotion path itself is proven on a synthetic rush hour fixture in `test/eta-train.test.ts`, where the model wins and is promoted.

Spine 1 arrival, leave one journey out (route HEIGHTS-REZENDE, real field ride)	Value
Recorded journeys / segment observations	2 / 20
Baseline, per segment average	96.4 s over 20 predictions
Model, per segment per hour average	96.4 s over 20 predictions
Verdict / served engine	<code>insufficient_data</code> / <code>baseline:v1</code>

The corridor dataset grows with the team's routine travel, and the arrival model promotes itself automatically the moment it clears the bar. Every rendered arrival says how many rides it stands on, so the estimate is always honest about its own evidence.

Spine 2: commute alerts, deliberately not a model

A recurring trip is a counting problem, not a learning problem, so Spine 2 trains no model, and that is the point. A plain miner groups the rider's own tickets from the last 28 days and calls a stop pair a pattern when it has at least 5 rides over at least 3 distinct weekdays (`apps/web/src/lib/commute/patterns.ts`). The named baseline is the fixed alarm clock every commute app ships: "your 07:45 kombi". The alarm knows the rider's habit but cannot know today's supply. Svika fires the alert only when Spine 1 says a real vehicle is actually near, so on a day the kombis run early or run sparse, the alarm lies and the alert stays honest. A fixed alarm cannot know today's supply; that sentence is the whole justification for the design, and it is why this spine is statistics, not machine learning.

Spine 3: revenue anomaly detection

An owner cannot watch four kombis' takings by eye across months. The baseline is a fixed threshold, the "60 percent drop" rule an owner might set themselves. The model is an isolation forest over per route, per day features. The forest serves only while it beats the threshold on F1 over held out labelled days (`services/spine/src/watchdog/eval.ts`). Over 360 held out days, of which 22 carry injected leakage, the forest scores an F1 of 0.756 at 73.9 percent precision and 77.3 percent recall, while the fixed threshold never fires and scores zero (`services/spine/metrics/WATCHDOG-METRICS.md`, generated by `pnpm watchdog:eval`).

The forest is promoted and serves. The threshold never fires because real leakage hides inside one kombi of four: a heavy skim on one vehicle moves the route total by well under the 40 percent drop the threshold needs. That dilution is the real world case, and it is exactly the argument for a model over a rule here. Every flag describes a pattern, a day, a route, an unnamed vehicle, never a person, and a unit test enforces that rule on the explanation templates.

Spine 3 watchdog, 360 held out days, 22 with injected leakage	Precision	Recall	F1
Fixed threshold, 60% drop rule (never fired)	0.0%	0.0%	0
Isolation forest, per route per day features	73.9%	77.3%	0.756

Honesty tiers, named on the record

Svika labels every feature by how real it is, in the README and a committed disclosure register, on the tier scale Section 4 defines. The three spines above are Tier 1: the detectors are real and their evaluations are real. Where they stand on generated history, Section 3 and the dataset statement say exactly what is real and what is simulated.

Section 3: Deliverables and CCE implementation roadmap

What is already built

Svika is not a concept. It is a working monorepo with a live database, passing tests and recorded demo evidence. The following are Tier 1, real and running against the live Supabase project, unless marked otherwise.

- **Identity and access.** Phone OTP sign in, roles derived from the network, sessions enforced under row level security. Consent gate on every surface before any personal data is touched, with an append only consent record and a plain language privacy notice.
- **The money and ticket core.** Append only double entry ledger, event sourced tickets, dated fare segments seeded at verified 2026 fares, a graph trip planner with transfer penalties proven over every stop pair, cash reservation and fare plausibility guards. Change becomes wallet credit through the ledger. Credit transfers, parcels and the owner ledger view are built and tested (phase 1 gate, 16 end to end tests, ledger 8 of 8).
- **The rider surface.** Search, plan and pay across Harare in Shona and English, a boarding card with a redemption stamp, the wallet, saved trips, share my ride as a tokenised live link, and commute alerts wired to the live arrival predictor.
- **The conductor surface.** An offline first installable app that clears fares with no signal, holds a local ticket cache, and reconciles queued writes on sync with first sync wins. Board codes are four digits, scoped to route, direction and time window, rate limited with attempt logging.
- **The live map and real data.** A MapLibre live map over the real Mount Pleasant Heights to Rezende corridor, seeded from the corridor rides the team records with a custom field GPS logger that turns routine travel into structured data. Arrival numbers are served by the real spine over real segment times, every estimate labelled with how many rides it stands on.
- **The AI spines.** All three built and evaluated with committed metrics: arrival prediction (baseline serving, promotion path proven on a fixture), commute alerts (statistics, live), revenue watchdog (isolation forest promoted, F1 0.756 against a threshold that never fires). See Section 2.
- **The owner surface.** Revenue dashboard, printable statement, the bilingual watchdog narrative, and the dated ZIMRA presumptive tax card. Watchdog history is labelled simulated on screen and in the register.
- **Staged, clearly labelled Tier 2 work.** A USSD menu state machine and a rendered feature phone for the no smartphone rider, and a staged crash alert flow. These are built behind tested state machines but move no real money and call no live telco; each carries an on screen caption saying so, and each awaits a partner agreement.

Discipline is visible in the history. Commits are small and conventional, the lockfile is committed, dependencies are pinned, and continuous integration runs typecheck, lint and test on every push. Current validation stands at 287 unit tests across the workspace, 56 full web end to end tests, and the 102 check security suite, all green, with the ledger invariants at 8 of 8 and the offline cycle at 34 of 34, all green in CI.

Every claim above is committed and inspectable. The code, migrations, tests, metrics and full commit history live in the public repository at github.com/takmaswi/svika-ai4i.

Compute requirements

The three models are deliberately light: an isolation forest and a pair of averaging predictors, CPU only, no GPU, no large weights, server side by product rule. That keeps the compute footprint small enough for a modest national compute allocation and removes the edge memory and latency risk heavier designs carry.

Deploying and testing in the ZCHPC CCE

The architecture was chosen with the national compute in mind. The spine service is a stateless container with a locked manifest; it takes ticket and ride data in and returns predictions and flags, holding no rider identity. That makes it a clean fit for the ZCHPC Cognitive Computing Environment: a single container to schedule, no special hardware, reproducible from the committed lockfiles.

The data layer runs today on managed Postgres in the vendor's eu-west-2 region, chosen for the most consistent round trip time to Harare because the vendor offers no African region. Moving the data in country is a roadmap item aligned with Pillar 2 of the National AI Strategy on computational sovereignty. The deployment plan for the challenge window is: containerise the spine and run it in the CCE against a staging database; measure inference latency and throughput on the allocation; then stand up an in country Postgres with the same migrations and row level security, and cut the data layer over. The migrations, security tests and seed are all reproducible, so this move is exercised, not improvised.

The same national compute is the intended home for a second roadmap deliverable: a purpose built Shona model trained on ZCHPC, replacing the machine drafted strings with real modern nuance, planned in full against low resource machine translation evidence in a committed roadmap plan and never claimed as built.

Milestones through the challenge window

The path is phase gated, and a phase closes only when its named proof exists, never on a calendar. Near term milestones, in order:

1. **Shona translator pass.** Every Shona string in the app is currently machine drafted placeholder text; an external Zimbabwean translator pass is owed before submission. This is a known, scoped gap, not a hidden one.
2. **More corridor ride days.** The corridor dataset grows with the team's routine travel through the same logger, and the arrival model promotes itself automatically the moment the data clears the promotion bar. A corridor ride day is planned with association contact.
3. **Voice guidance with consent.** Zimbabwean voice artists sign consent, and the in ride guidance audio is pre generated and cached so no vendor sits in the ride path. A placeholder voice ships if consent is not yet secured.
4. **CCE deployment test.** Containerise and run the spine in the ZCHPC CCE, measure latency and throughput, then rehearse the in country data cutover described above.
5. **Bootcamp and pilot readiness.** The Track 3 bootcamp runs in Mutare from 27 July to 1 August 2026. Svika enters it with a working product and a security suite, ready for milestone based support toward a corridor pilot.

Section 4: Compliance and risk mitigation

Data protection: built to the Act, licensed before live processing

Svika is designed against the Cyber and Data Protection Act [Chapter 12:07] and its licensing regulations, Statutory Instrument 155 of 2024, which POTRAZ administers. The judge reading this section is the regulator, so the claims here are specific and checkable.

Consent is a gate, not a checkbox at the bottom of a form. No Svika surface touches personal data until the rider or conductor passes a first use consent gate, and the conductor gate works offline against a cached verdict. Consent is stored as an append only `consent_records` row under row level security, alongside a plain language privacy notice that lists exactly what Svika holds: profile, tickets, wallet ledger, saved trips. A rider can withdraw. The "what Svika knows about you" page runs an `anonymise_me` function that strips name and phone and deletes saved trips; because the ledger and tickets are append only by law, the page says plainly that history is anonymised rather than pretending full erasure. All of this is built and tested, with 8 consent checks inside the security suite.

Licensing is planned honestly for the point it is needed. SI 155 requires any entity processing the personal data of 50 or more individuals to hold a data controller licence and to appoint a Data Protection Officer within 90 days. Today Svika processes only the team's own data and synthetic demo personas; it is not yet processing real riders at scale. The compliance plan is to register as a Tier 1 data controller, the USD 50 tier under SI 155, and to appoint a Data Protection Officer before the first real rider is onboarded, not before it is required. Naming this now, rather than claiming a licence we do not yet need, is the honest position and the one the Act actually expects.

Breach notification is a documented procedure. The Act requires a data controller to notify the Authority within 24 hours of a security breach, and to inform affected individuals within 72 hours where the risk to their rights is high. Svika's operating plan adopts both windows as written, and the appended consent and audit records give the factual trail a notification would need.

Cybersecurity: integrity enforced in the database

The strongest security control in Svika is structural, not a bolt on. Row level security is on every table from the first migration, so authorisation is enforced by Postgres on every query, not by application code that a bug could bypass. An automated test proves rider isolation using the anonymous key alone (102 checks passing). The service role key exists only in the seed script and CI secrets and never in a running app. No secret is committed to the repository; `.env.local` is git ignored and `.env.example` documents every key. This directly answers the terms of reference disqualification flag on committing credentials, and the C1 requirement for code free of security warnings: the database advisor runs clean of actionable items.

Board codes are hardened by design: four digits, scoped to a route, direction and time window, with rate limited redemption and attempt logging, so a guessed or shared code cannot travel across the network. Fare purchase carries a plausibility guard and cash reservation. Every external provider sits behind an adapter with a mock twin, so no live vendor call sits in the ride path and the demo cannot die because a vendor is down.

Model testing and AI risk

Each of the three AI spines ships with unit tests and a committed metrics file, so the model's behaviour is evidenced, not asserted (Section 2). Two AI specific risks get named product rules. First, forced AI: where a rule or query is sufficient, Svika uses it and says so, which is why the commute alert is statistics rather than a model. Second, accusation: the revenue

watchdog flags a pattern, a day, a route, an unnamed vehicle, and never a person, and a unit test enforces that on the explanation templates. An owner sees "one vehicle on this route ran low on this day", never a named driver accused of theft.

Honesty and adoption risk

Svika labels every feature by honesty tier, in the README and the disclosure register: Tier 1 real, Tier 2 clickable with a fixed backend and labelled on screen, Tier 3 slides only. We never present a Tier 2 surface to a judge as live. This is a deliberate guard against the terms of reference weak submission flags about polished screenshots with no working path, and against overclaiming AI performance. The dataset statement (Section 3 and C3) carries the same discipline for data: what is real, what is simulated, and how the simulator will be checked against real aggregates as they arrive.

Adoption risks are real and stated rather than hidden. Cash is always accepted and digital riders get no queue privilege, product law carried over from the ZUPCO post mortem, so Svika adds a record without punishing the rider who pays cash. The offline first conductor design assumes a cheap Android on a weak network, because that is the real device. The Shona in the app is machine drafted placeholder pending an external translator pass, which is flagged, not glossed over. None of these are fatal, and each has an owner and a plan.

Section 5: Sustainability and future adoption

The business model: sell the record, not the ride

Svika does not tax the rider to survive. The paying users are the owners and, through them, the network, because Svika produces something they cannot get today: a trustworthy record of what each kombi earns and carries.

Owner subscription. Kombi owners pay a monthly presumptive tax set by ZIMRA Public Notice 51 of 2025, gazetted on 5 September 2025 and collected through the ZINARA licence: USD 50 a month for an 8 to 14 seater and USD 60 for a 15 to 24 seater, which is the kombi range, as a fixed amount rather than a share of takings. Since that notice an operator cannot renew the licence without a tax clearance, so the earnings record an owner brings to ZIMRA now decides whether the vehicle stays on the road. Today an owner meets that demand with no evidence of what a vehicle really earned. Svika's dashboard, printable statement and revenue watchdog give the owner a digital earnings record beside the fixed tax, with an isolation forest watching for leakage the eye cannot catch. That is worth a monthly fee per vehicle, and it is the primary revenue line: a subscription priced well under the value of the leakage it surfaces and the credibility it gives the owner with the tax authority.

Conductor commission and the wallet float. Conductors earn a commission on digital fares, carried in the ledger as a per fare rate (`commission_rate_bps`), which gives the crew a direct reason to clear fares digitally rather than route around the system. Unreturned change becomes USD wallet credit through the same double entry ledger; the aggregate float is a second, smaller economic base and, more importantly, the hook that keeps riders in the digital record.

Cost structure

The cost side is deliberately light, which is what makes a per vehicle subscription sustainable at Zimbabwean prices.

- **Compute.** The three models are CPU only, with no GPU and no large weights (Section 2). Server side inference on a modest allocation covers the network, and the ZCHPC national compute is the intended long term home, which lowers hosting cost further.

- **Data layer.** Managed Postgres on an entry tier today, with an in country move on the roadmap; the load is transactional and small per vehicle.
- **Display.** Map tiles are a licensed MapTiler service used only for display, a fixed service cost that does not scale with rider count.
- **No per ride vendor cost.** Voice audio is pre generated and cached, and every external provider sits behind an adapter, so there is no metered vendor call in the ride path to erode the margin. [FLAG: specific monthly infrastructure figures are estimates pending a costed hosting plan; see traceability note.]

The unit economics are simple: revenue is per vehicle per month, the marginal cost of one more vehicle is close to zero, and the heavy costs are fixed and low. That is a model that survives on a few hundred kombis and improves with scale.

Edge posture and resilience

Svika is server first by product rule, and that answers the edge question rather than dodging it. The models run on CPU, so no rider or conductor phone runs inference, and riders' data stays off the device and in country on the ZCHPC national compute, the data sovereignty position Pillar 2 of the National AI Strategy asks for. The real edge risk on a kombi is losing signal in traffic, and that is met where it bites, by the offline first conductor app that clears fares and holds a local ticket cache with no network then reconciles on sync, not by a quantised model on a cheap phone.

Licensing register

Svika keeps a maintained record of what it depends on and under what right, which is the licensing registry the terms of reference ask for. The field GPS traces, stop names and the measured fare were collected by the team and sit under the repository licence, publicly visible for adjudication. The synthetic histories are generated by committed code under the same licence. No third party dataset is used, and no dataset holds a real rider's personal data. The planned Shona training corpus, a paid human Golden Seed multiplied by filtered synthetic generation and evaluated on the professional FLORES-200 held out set, is documented as roadmap in a committed plan; it is future data, not yet collected, and is labelled as such in the dataset statement. On the regulatory side, Svika will hold the Tier 1 data controller licence under SI 155 before live processing (Section 4). The register is updated in the same commit as any change to a dependency or data source.

Future adoption

Three roadmap lines extend the base without changing the promise. First, a Kombi Trust Value: as a vehicle builds a clean digital record of trips carried and fares cleared, that record becomes a portable trust signal an owner can show a lender, an insurer or an association. Second, the voice guidance grows from a safety and accessibility feature into a Shona language tool the wider network can license, matching the National AI Strategy's commitment to natural language processing in local languages. Third, anonymised and aggregated network data, never a rider's personal data, can inform city transport planning of the kind Harare's own master plan calls for, the same public value that turned Nairobi's matatu traces into that city's first complete transit map; a municipality or planning agency is a natural second customer for that aggregate insight, and it is roadmap, not a line we bill today. Each of these is roadmap, clearly labelled, and each earns its place only after the core record keeping product is trusted and paid for.